An explicit quartic construction over the rationals

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Abstract

We give a complete quartic coefficient table in 456 total variables, obtained from an explicit quadratic arithmetic system. The algebraic transformations, the local normalization, the index-depth calculation and the coefficient-height formula are proved. The rational input projection is the integers under four stated source arithmetic inputs and an explicitly specified weighted prime-selection input. The polynomial has primitive integer coefficients and height $64 \cdot 3^{105665812}$. Complete data and independent reconstruction checks accompany the paper.

Contents

1 An explicit quartic and its meaning **2**

2 Arithmetic inputs and the product-height identity **2**

3 The exact numerical height constant **5**

4 Simultaneous parity and divisibility conditions **5**

5 Prescribing primary divisibility with the existing anchor **7**

6 Exact three-adic depth of the anchor **10**

7 A scalar normalization without squareclass corrections **13**

8 Asymmetric formal depth for the quartic **16**

9 Quadratic circuits and the explicit quartic **19**

10 The input circuit and fixed coefficients **21**

11 A quartic in 456 total variables **23**

12 The complete coefficient table **26**

1 An explicit quartic and its meaning

A polynomial $Q \in \mathbb{Z}[T, Y_1, \dots, Y_m]$ has rational input projection $\{t \in \mathbb{Q} : \exists \mathbf{y} \in \mathbb{Q}^m, Q(t, \mathbf{y}) = 0\}$. All variable counts include the input T. The coefficient height is the largest absolute coefficient.

The supplied polynomial is the sum of the squares of a fully listed quadratic system. Its coefficient table, degree, primitivity and height are exact algebraic assertions. The equality of its input projection with $\mathbb{Z}$ uses the arithmetic inputs of Section 2 and the weighted prime-selection input of Section 8. The constructions below prove the transformations and their soundness and completeness on the stated domains; the source arithmetic premises remain explicit.

The starting arithmetic system, its local extraction argument and the simultaneous approximation construction are given in [1, 2]. The present paper develops the quadratic implementation, its exact local index conditions, and the transformations needed for the supplied coefficient table.

2 Arithmetic inputs and the product-height identity

2.1 Arithmetic inputs and conventions

Put

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and fix

$$R_0 = (1 + \sqrt{2}, 2\sqrt{2}), \qquad P = 2R_0 = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

Write σ for conjugation, $T_0 = (0, 0)$, $f(Q) = 1/x(Q)$, and $z(Q) = -x(Q)/y(Q)$. Heights are absolute logarithmic Weil heights, with normalized local degree weights. Valuations at unramified primes satisfy $v(p) = 1$.

We use the following arithmetic hypotheses, with the named source locations in [2].

1. (Elliptic group and height.)
   Source: `subsec:v5-point-source`, `eq:o8-indexheight`.
   $2E(K) = \mathbb{Z}P$, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.

2. (Contact height: Theorem `thm:quartic-height19`.) For $t = u/v$ in lowest terms, $v > 0$, and $t \notin \mathcal{D} = \{0, 1, 2401, -343/32\}$, define $M_5(t)$ to be the product of the primes $p > 76176$ for which at least one of

   $$v_p(v), \quad v_p(u), \quad v_p(u - v), \quad v_p(u - 2401v), \quad v_p(32u + 343v)$$

   is positive and odd. These are the normalized contacts with $\infty, 0, 1, 2401, -343/32$. Put

   $$B_* = \prod_{p \leq 76176} p, \qquad H = H_{19} = 2^{110} 210^{1800} B_*^{19} + 128.$$

   The inherited assertion is $h(t) \leq H M_5(t)^{19}$. Exact integer evaluation gives $H < 2^{2093448}$.

3. (Pole obstruction and realization.)
   Source: `hyp:o8-pole` and `lem:original-seven-simultaneous-realization`. The pole equations, with $B = 17b$,

   $$U = b\frac{1 - bt^2}{1 + bt^2}, \quad h_0 = e(B - U^2), \quad p^2 = B(1-e)(1+3e), \quad Bq^2 = (s^2 - Bh_0)(s^2 + 3Bh_0),$$

   exclude negative odd $v_\ell(b)$ for $\ell \notin \{2, 3, 17\}$. They are soluble with $t = 0$ whenever b has square reduced denominator, is positive and less than 17, is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and satisfies $v_2(b) \geq 3$, $v_3(b) \geq 1$, $v_{17}(b) \geq 2$.

4. (Simultaneous prime patterns: Theorem `thm:v5-simultaneous-primes`, Lemma `lem:v5-two-reductions`, and the application of [3, Theorem 4.1].) The simultaneous Gaussian prime-pattern input supplies the prescribed prime values $u + c_j w^2$, arbitrary fixed divisibility of w, and arbitrarily large u/w^2. Their two conjugate reduction orders satisfy

$$v_2(d_-) = 6, \quad v_2(d_+) = 2, \quad \gcd(d_-(p), d_+(p')) = 4$$

for every pair of selected primes. An arbitrary fixed odd avoidance modulus can be imposed. For each selected Gaussian prime π, the source convention is $\pi \equiv 1+3i \pmod 7$, making both $N(\pi - 1)$ and $N(\pi + 1)$ units at seven. At each other odd prime ℓ in the fixed avoidance modulus, the condition $\pi \equiv i \pmod \ell$ makes both orders congruent to two modulo ℓ. Thus the orders avoid the entire odd part of the fixed modulus. The selected rational primes are 1 modulo $8 \cdot 3 \cdot 17$.

These hypotheses include the simultaneous application of Sun's stated prime-pattern theorem. They are assumptions of the theorem below, not conclusions of its implementation audit. The local depth of the displayed point is proved in Proposition 6.1.

2.2 An elementary Lucas growth estimate

The following estimate is independent of the arithmetic hypotheses. It supplies the growth inequality used in the integer-extraction argument.

Lemma 2.1. *Let $a, t, b \in \mathbb{Z}$, with $|a| > 7$ and $b \neq 0$, satisfy*

$$t^2 - a(a-2)tb + (a-2)^2 b^2 = 1.$$

Then $\log|t| > |a| - 3$. *In particular,* $\log\max(|a|, |t|, |b|) > |a| - 3$.

Proof. Put $A = |a|$, $c = (a-2)b$ and $c' = \operatorname{sgn}(a)c$. Then

$$t^2 - Atc' + (c')^2 = 1.$$

We have $|c'| \geq |a-2| \geq 6$, so $t \neq 0$. The two terms t, c' have the same sign: opposite signs would make the left-hand side at least $t^2 + (c')^2 > 1$. Thus $T = |t|$ and $C = |c'|$ are positive integer solutions of $T^2 - ATC + C^2 = 1$.

Define $U_0 = 0$, $U_1 = 1$ and $U_{j+1} = AU_j - U_{j-1}$. Every positive solution consists of adjacent terms of this sequence. Indeed, order its coordinates as $M \geq m > 0$. The integer

$$Am - M = \frac{m^2 - 1}{M}$$

lies in $[0, m)$, and replacing (M, m) by $(m, Am - M)$ preserves the equation. Repeated descent ends at $(1, 0)$. Reversing the steps gives the unordered pairs $\{U_{j+1}, U_j\}$. Consequently $C = U_j$ for some $j \geq 1$ and $T \geq U_{j-1}$.

Put $d = |a-2|$. Since $d \mid C$, reduction of the recurrence modulo d gives $d \mid j$: if $a > 0$, then $A \equiv 2 \pmod d$ and $U_j \equiv j \pmod d$; if $a < 0$, then $A \equiv -2 \pmod d$ and $U_j \equiv (-1)^{j-1} j \pmod d$. Hence $j \geq d \geq A - 2$. The recurrence also gives $U_r > (A-1)^{r-1}$ for $r \geq 2$. This follows by induction from $U_2 = A > A - 1$ and $U_{r+1} > (A-1)U_r$, since $U_r > U_{r-1}$. As $j \geq 6$, we obtain

$$\log T > (j-2)\log(A-1) \geq (A-4)\log 7 > A - 3 \qquad (A \geq 8).$$

The last inequality holds at $A = 8$ and its difference increases with A. This proves the assertion. □

2.3 An exact height identity

Lemma 2.2. *For any nonzero odd integers $n_1, \dots, n_r$,*

$$h\left(\prod_{i=1}^{r} x([n_i]P)\right) = \sum_{i=1}^{r} h(x([n_i]P)).$$

This identity requires only that the displayed $P = 2R_0$ have infinite order; it does not require the contact-height or prime-pattern hypotheses.

Proof. First consider a finite place v with $v(D) = 0$. The formal subgroup $E_1(K_v)$ consists of the identity and points with $v(x) < 0$. It is an open subgroup, so $E(K_v)/E_1(K_v)$ is finite. These facts hold for the displayed integral Weierstrass equation, including the place above two. The image of T_0 has order two, since $T_0 \notin E_1(K_v)$, and

$$x(Q + T_0) = -D/x(Q).$$

If an odd multiple of P belongs to $E_1(K_v)$, the image of P in the quotient has odd order. No multiple of that image is the nonzero element represented by T_0. Hence no odd multiple of P has positive x-valuation. If no odd multiple belongs to $E_1(K_v)$, all odd multiples have nonnegative x-valuation. Thus the numbers $v(x([n_i]P))$ have a common weak sign.

The only remaining finite place is the prime $(3 - \sqrt{2})$ above seven. Here $v(D) = 2$, and the discriminant has valuation six, so the displayed integral model is minimal. Its reduction is $y^2 = x^3$. The displayed P reduces to $(1, 6)$, a nonsingular point. The subgroup of points with nonsingular reduction therefore contains every multiple of P. A point with positive x-valuation would have positive y-valuation and reduce to the singular point $(0, 0)$. Consequently every multiple of P has x-valuation at most zero at this place.

At either real embedding every $[n_i]P$ is a nonzero double. The duplication identity

$$x(2Q) = \left(\frac{x(Q)^2 + D}{2y(Q)}\right)^2$$

makes its abscissa positive. The curve equation and its nonzero ordinate imply $x([n_i]P) > d > 1$ in each embedding. Each real local height therefore adds under multiplication. At a finite place the common weak sign similarly gives

$$\max\{0, -\textstyle\sum_i v(x([n_i]P))\} = \sum_i \max\{0, -v(x([n_i]P))\}.$$

Summing with the normalized local weights proves the identity. □

Corollary 2.3. *Let $Q_1, \dots, Q_4$ be odd multiples of P and write*

$$C = \prod_{i=1}^{4} x(Q_i) = C_0 + C_1\sqrt{2}.$$

The three numbers $A_0 = 1$, $A_1 = C_0^2$, and $A_2 = (1 + C_1^2)^2$ are nonzero positive rational squares. If the three contact detectors at $A_j\lambda$ imply $M_0(A_j\lambda)^k \mid N \neq 0$, then

$$h(x(Q_i)) < 7H|N|^{19/k} \qquad (1 \le i \le 4).$$

Here M_0 is the odd-contact radical for $\Theta = \{0, -1/2, -1, 1/2399, -32/407\}$.

Proof. The product is positive in both embeddings, so $C_0 > 0$. The fractional linear map $t \mapsto 2 + 1/t$ identifies the five contacts with the original five, outside the exceptional primes. Consequently $h(t) \le (H + \log 3)M_0(t)^{19}$. Put $T = (H + \log 3)|N|^{19/k}$. The unscaled detector and each scaled detector give $h(A_j) \le 2T$. Hence $h(C_0) \le T$. For $C_1 = u/v$ in lowest terms with $v > 0$, $h(1 + C_1^2) = \log(u^2 + v^2) \ge 2h(C_1)$, so $h(C_1) \le T/2$. The elementary height inequality in the fixed basis gives

$$h(C) \le \tfrac{3}{2}T + \tfrac{3}{2}\log 2 < 7H|N|^{19/k}.$$

Lemma 2.2 bounds every summand by $h(C)$. □

3 The exact numerical height constant

The contact-height input uses a fixed, explicitly specified integer. Its exact evaluation improves the numerical bounds in both construction methods. Put

$$B_* = \prod_{p \le 76176} p, \qquad H_{19} = 2^{110} 210^{1800} B_*^{19} + 128.$$

This is the constant of the source's exponent-nineteen height theorem [2, Theorem `thm: quartic-height19`].

Lemma 3.1. *The exact inequalities*

$$2^{2093447} < H_{19} < 2^{2093448}$$

hold. In particular one may use $h = 2093448$ in every argument requiring $H_{19} < 2^h$.

Proof. This finite integer calculation has the following reproducible certificate. Enumerate primes through 76176 by marking, for each unmarked $p \le \lfloor \sqrt{76176} \rfloor$, all multiples starting at p^2. The remaining integers at least two are exactly the primes: every composite has a prime divisor not exceeding its square root, and no prime is marked. An independent enumeration tests every integer by trial division through its integer square root. Both lists have 7499 entries and agree entry by entry.

Their exact product has 109445 binary digits. Computing the displayed powers and product as integers gives 2093448 binary digits for H_{19}. The comparison with the next smaller power of two is strict as well. The supplied checker performs both prime enumerations and both integer comparisons; the complete prime list and the unsigned binary encoding of H_{19} are included. No floating-point logarithm enters these comparisons. □

This lemma evaluates a constant. It does not prove the contact-height inequality itself. The latter is still the arithmetic input stated in Section 2, now used with this exact value and its sharper certified bound.

4 Simultaneous parity and divisibility conditions

Put $K = \mathbb{Q}(\rho)$, where $\rho^2 = 2$, and consider

$$d = 3 - \rho, \qquad D = d^2 = 11 - 6\rho, \qquad E:\ y^2 = x^3 - Dx, \qquad P = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32} \right).$$

We write v_p for the rational valuation with $v_p(p) = 1$. The valuation ν_2 on K above 2 is normalized by $\nu_2(\rho) = 1$, so $\nu_2(2) = 2$. Let $R = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$, and set

$$N(a, b) = a^2 + ab - b^2.$$

Lemma 4.1. *For rational a, b, c, e,*

$$N(a, b) + 6N(c, e) \in R \quad \Longleftrightarrow \quad a, b, c, e \in R.$$

Proof. The binary form N has no nontrivial zero over either $\mathbb{F}_2$ or $\mathbb{F}_3$. Consequently, for $p = 2, 3$ and $(a, b) \neq (0, 0)$,

$$v_p(N(a, b)) = 2 \min(v_p(a), v_p(b)).$$

Indeed, divide both inputs by the smaller power of p and reduce modulo p. The two summands $N(a, b)$ and $6N(c, e)$ have valuations of opposite parity when both are nonzero. They therefore cannot cancel. Their sum is p-integral if and only if

$$2 \min(v_p(a), v_p(b)) \ge 0, \qquad 1 + 2 \min(v_p(c), v_p(e)) \ge 0.$$

Since these minima are integers, these inequalities say that all four inputs are p-integral. If either pair is zero, the same conclusion follows by omitting the corresponding zero term. □

Lemma 4.2. *For $t \in \mathbb{Q}$,*

$$t \in R \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q} : \; u^2 + 6v^2 - 2w^2 - 3t^2 + 2 = 0.$$

Proof. This is the three-witness semilocal criterion used by the source construction; we recall its arithmetic justification. The Hilbert-symbol formulas show that $(2, 3)_{\mathbb{Q}}$ is ramified precisely at 2 and 3: outside $2, 3$ both parameters are units and the algebra splits; at 3 the symbol is the Legendre symbol $(2/3) = -1$, and reciprocity gives the symbol -1 at 2 since the real place is split. Its quadratic-field embedding criterion [4, Proposition 14.6.7] says that a nonzero rational number Δ is the square of a trace-zero element precisely when Δ is nonsquare in both completions. For a rational square $\Delta \neq 0$, such an element is impossible in a division algebra: $(q - \sqrt{\Delta})(q + \sqrt{\Delta}) = 0$ would make q scalar. For nonsquare Δ, this is the usual local embedding criterion for quadratic fields in quaternion algebras.

For $t \neq 0$, take $\Delta = 1 - 2/(3t^2)$, which is nonzero because $2/3$ is not a rational square. The trace-zero square form is $2x^2 + 3y^2 - 6z^2$. At 2, if $v_2(t) > 0$, then $v_2(\Delta)$ is odd; if $v_2(t) = 0$, then $\Delta \equiv 3 \pmod 8$; and if $v_2(t) < 0$, then $\Delta \in 1 + 8\mathbb{Z}_2$ is a square. At 3, a nonnegative valuation of t gives odd negative valuation of Δ, whereas a negative valuation gives a square in $1 + 3\mathbb{Z}_3$. Hence the embedding criterion holds exactly when $t \in R$. Multiplying $2x^2 + 3y^2 - 6z^2 = \Delta$ by $3t^2$, and putting $v = tx$, $u = 3ty$, $w = 3tz$, gives the displayed equation. The case $t = 0$ has witnesses $(u, v, w) = (0, 0, 1)$. □

Lemma 4.3. *For every nonzero integer n,*

$$\nu_2(x([n]P)) = -6 - 4v_2(n).$$

At the place above 3, every odd multiple $[n]P$ has unit abscissa.

Proof. The displayed coordinates give $\nu_2(x(P)) = -6$, $\nu_2(y(P)) = -9$, and therefore $\nu_2(z(P)) = 3$ for the formal parameter $z = -x/y$. The equation is integral at 2. Its formal logarithm is an isometry on the subgroup of depth strictly greater than $\nu_2(2)/(2-1) = 2$. No minimal-model assertion is needed for this integral formal group [5, Theorem IV.6.4]. Its nonzero logarithm also proves that P has infinite order. Since the logarithm is a homomorphism,

$$\nu_2(z([n]P)) = 3 + 2v_2(n).$$

The Laurent expansion $x = z^{-2}(1 + O(z))$ then proves the formula.

At 3 the equation has good reduction: its discriminant is $64D^3$, a unit. The residue field is $\mathbb{F}_9$, and direct reduction gives $\overline{P} = (\overline{\rho}, 0)$. This point has order two and nonzero abscissa. Every odd multiple therefore has the same reduction and unit abscissa. □

Theorem 4.4. *Let $n_1, \ldots, n_r$ be nonzero integers and write*

$$C = \prod_{i=1}^{r} x([n_i]P) = C_0 + C_1\rho.$$

Then

$$n_1, \ldots, n_r \text{ are all odd} \quad \Longleftrightarrow \quad 2^{3r}C_0,\ 2^{3r}C_1 \in R.$$

Proof. The preceding lemma gives

$$\nu_2(2^{3r}C) = -4\sum_{i=1}^{r} v_2(n_i).$$

Each summand on the right is nonnegative. Thus this expression is nonnegative exactly when every index is odd. The local integer rings at 2 and 3 are respectively $\mathbb{Z}_2[\rho]$ and $\mathbb{Z}_3[\rho]$. At 2 this can also be seen directly from

$$\nu_2(a+b\rho) = \min(2v_2(a), 2v_2(b)+1),$$

whose two entries have opposite parity. At 3, the integral basis $(1, \rho)$ remains linearly independent modulo 3. Therefore integrality in either local ring is equivalent to integrality of both rational coefficients. If the indices are odd, the preceding lemma makes C a 3-adic unit, so the additional 3-integrality conditions impose no restriction. The converse already follows from the place above 2. □

5 Prescribing primary divisibility with the existing anchor

This section concerns rational points on one fixed curve. Its local statements do not use the prime-pattern or height inputs of the Diophantine construction. Those arithmetic inputs enter only when the construction supplies the specified cyclic point domain and its integer-input witnesses.

Put

$$K = \mathbb{Q}(\sqrt{2}), \qquad D = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and

$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

At the place above two, normalize the valuation ν_2 by $\nu_2(\sqrt{2}) = 1$, so $\nu_2(q) = 2v_2(q)$ for $q \in \mathbb{Q}^*$. Write

$$R = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}, \qquad N(a,b) = a^2 + ab - b^2.$$

All points whose reciprocal abscissas occur below are finite and nonidentity. All cyclic indices are nonzero integers.

Lemma 5.1. *For every nonzero integer n, write $f([n]P) = 1/x([n]P) = a_n + b_n\sqrt{2}$. Then*

$$\nu_2(x([n]P)) = -6 - 4v_2(n), \qquad v_2(a_n) = 3 + 2v_2(n), \qquad v_2(b_n) \geq 3 + 2v_2(n).$$

At the place above three, an odd multiple of P has unit abscissa, whereas an even nonzero multiple is formal and its reciprocal abscissa has both rational coefficients in $\mathbb{Z}_3$. For $p = 2, 3$ and rational a, b, not both zero, one has

$$v_p(N(a,b)) = 2\min(v_p(a), v_p(b)).$$

Proof. The equation for E is integral at two. The displayed coordinates give $\nu_2(x(P)) = -6$ and $\nu_2(y(P)) = -9$. Thus the formal parameter $z = -x/y$ has $\nu_2(z(P)) = 3$. The formal logarithm is an isometry when $\nu_2(z) > 2$, since the absolute ramification index is two. Multiplication by n therefore gives

$$\nu_2(z([n]P)) = 3 + 2v_2(n), \qquad \nu_2(x([n]P)) = -2\nu_2(z([n]P)).$$

The logarithm also shows that these multiples are nonidentity. For rational a, b, the two quantities $2v_2(a)$ and $2v_2(b)+1$ have different parity. Hence

$$\nu_2(a + b\sqrt{2}) = \min\{2v_2(a), 2v_2(b)+1\}.$$

Applied to $f([n]P)$, whose valuation is the even integer $6 + 4v_2(n)$, this proves the two coefficient assertions.

At three the curve has good reduction and $\overline{P} = (\sqrt{2}, 0)$ has order two. Its abscissa is a unit. This proves the assertions for odd and even multiples. The extension $\mathbb{Q}_3(\sqrt{2})/\mathbb{Q}_3$ is unramified, with integral basis $(1, \sqrt{2})$, so integrality of an element implies integrality of both displayed coefficients.

Finally, $X^2 + XY - Y^2$ has no nonzero zero over either $\mathbb{F}_2$ or $\mathbb{F}_3$. After factoring the common minimum power of p from a and b, reduction modulo p therefore gives a unit. This proves the norm valuation formula. □

Lemma 5.2. *Let $1 \le s \le 7$ and $r \ge 1$. For points $U_i = [u_i]P$, write $f(U_i) = a_i + b_i\sqrt{2}$ and put*

$$T_r = 2^{-(4r+5)} \sum_{i=1}^{s} a_i^2.$$

Let $m = \min_i v_2(u_i)$, and let k be the number of indices attaining this minimum. Then

$$v_2(T_r) = 1 + 4(m - r) + v_2(k).$$

Consequently T_r is 2-integral if and only if $2^r \mid u_i$ for every i. Whenever T_r is not integral, its valuation is congruent to 1, 2, or 3 modulo four.

Proof. By Lemma 5.1, the numbers $w_i = a_i/2^{3+2m}$ are 2-integral. Exactly k of them are units, and every other w_i is divisible by four. A 2-adic unit square is congruent to one modulo eight, so

$$\sum_i w_i^2 \equiv k \pmod 8.$$

Since $1 \le k \le 7$, one has $v_2(k) \le 2$, and the displayed congruence determines the valuation of this sum exactly. Rescaling gives

$$v_2(T_r) = 6 + 4m - (4r + 5) + v_2(k) = 1 + 4(m - r) + v_2(k).$$

This is positive for $m \ge r$ and negative for $m < r$. Its residue modulo four belongs to $\{1, 2, 3\}$. The sum is nonzero, also directly because its rational summands are squares and every a_i is nonzero. □

The bound of seven is necessary for this particular unweighted test. For eight identical points $U_i = P$ and $r = 1$, one has $T_1 = a_1^2/64$ and $v_2(T_1) = 0$, although all eight indices are odd.

Theorem 5.3. *Let $Q_j = [n_j]P$, $1 \le j \le 4$, be four nonzero cyclic points, and write*

$$C = \prod_{j=1}^{4} x(Q_j) = C_0 + C_1\sqrt{2}, \qquad B = 2^{24} N(C_0, C_1).$$

Let $S = [a]P$, with $a \ne 1$, and write $f(S - P) = f_0 + f_1\sqrt{2}$. Fix a nonnegative integer q and put $C_* = 3^q$, $S_0 = C_*^2$, and

$$A = \frac{N(f_0, f_1)}{S_0}.$$

With $1 \le s \le 7$ primary points and T_r as in Lemma 5.2, the condition

$$A + B + T_r \in R$$

is equivalent, on the stated cyclic point domain, to all three of the following conditions:

$$\frac{f_0}{C_*}, \frac{f_1}{C_*} \in R, \qquad n_j \text{ is odd for every } j, \qquad 2^r \mid u_i \text{ for every } i.$$

Proof. Set $M = \sum_j v_2(n_j)$. Lemma 5.1 gives

$$\nu_2(2^{12}C) = -4M.$$

The valuation is even. Therefore $v_2(2^{12}C_0) = -2M$ and $v_2(2^{12}C_1) \geq -2M$; the norm valuation formula yields

$$v_2(B) = -4M.$$

The point $S - P = [a-1]P$ has nonzero integer index. Both coefficients of its reciprocal abscissa are 2-integral, and C_* is a 2-adic unit. Thus A is always 2-integral.

If B is not 2-integral, its valuation is a negative multiple of four. If T_r is not 2-integral, its valuation is not a multiple of four. Two such negative valuations are different, so their minimum is the valuation of their sum. If only one of B, T_r is nonintegral, its negative valuation likewise cannot be canceled by the other. Adding the integral term A cannot change either conclusion. Hence

$$A + B + T_r \in \mathbb{Z}_2 \quad \Longleftrightarrow \quad B, T_r \in \mathbb{Z}_2.$$

The right side says exactly that all n_j are odd and all u_i are divisible by 2^r.

Assume these conclusions. At three, each root point reduces to $(\sqrt{2}, 0)$, so C_0, C_1 and B are 3-integral. Since $r \geq 1$, every primary point is an even multiple and reduces to the identity. Its reciprocal abscissa has integral rational coefficients, and the denominator of T_r is a power of two. Thus T_r is also 3-integral. Subtraction now gives

$$A + B + T_r \in \mathbb{Z}_3 \quad \Longleftrightarrow \quad A \in \mathbb{Z}_3.$$

The norm valuation formula identifies the last condition with the simultaneous 3-integrality of f_0/C_* and f_1/C_*. Their 2-integrality was automatic. Conversely, the three stated conditions put each of A, B, T_r in R. □

Quadratic implementation. The theorem uses the single physical-anchor value

$$G = N(f_0, f_1) + S_0 \left(2^{24} N(C_0, C_1) + 2^{-(4r+5)} \sum_{i=1}^{s} a_i^2 \right).$$

This is a quadratic expression in coordinates already present. The existing ternary semilocal representation imposes $G/S_0 \in R$ with its existing three witnesses. No additional coordinate is needed. For $r = 1$, an invertible rational rescaling $B_j = 2^{12} C_j$ and $A_i = a_i/32$ writes the physical-value equation as

$$N(f_0, f_1) + S_0 \left(N(B_0, B_1) + 2 \sum_i A_i^2 \right) - G = 0.$$

The coordinate rescalings preserve the degree after fixed denominators are cleared.

Corollary 5.4. *In the construction with five primary points $U_0, \ldots, U_4$, take $r = 1$ and $Q_0 = P$, with $Q_1, \ldots, Q_4$ the four root points. On the strengthened point domain of Theorem 5.3, the two rational coordinates for each inverse $(x(U_i) - x(Q_i))^{-1}$ can be deleted. If these ten coordinates occur only in their inverse equations, their deletion changes neither the solutions of this strengthened system nor its input projection.*

Proof. The theorem makes each U_i an even multiple of P and each Q_i an odd multiple. Equality of their abscissas would imply $U_i = Q_i$ or $U_i = -Q_i$ on the short Weierstrass curve. The infinite order of P would then identify an even integer with an odd one, which is impossible. Each difference is therefore nonzero. The omitted inverse exists uniquely in K, giving its two rational coordinates and uniquely extending every solution of the reduced system. In particular, the retained slope equations have the same intended values. □

The comparison with the earlier construction requires a separate completeness argument. Its integer-input witnesses use primary indices $u_i = hn_i$ with $64 \mid h$; all five are already even. They therefore satisfy the strengthened domain and the new merged anchor. Under the inherited arithmetic hypotheses this preserves completeness at every integer input, while every solution of the strengthened system remains a valid solution of the earlier one. It is not asserted that every auxiliary solution of the earlier system has even primary indices. The ten-coordinate saving uses this proved source restriction, together with the independently checked occurrence condition on the deleted inverse coordinates.

6 Exact three-adic depth of the anchor

The coefficient size of the physical anchor can be reduced by using its exact local depth. The following calculation concerns only the fixed point and its integral multiples; it is independent of the prime-pattern and global height hypotheses.

Put

$$K = \mathbb{Q}(\sqrt{2}), \qquad D = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and

$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

The prime three is inert in K. Let v be the valuation on $K_3 = \mathbb{Q}_3(\sqrt{2})$ normalized by $v(3) = 1$. As usual, the valuation of zero is $+\infty$.

Proposition 6.1. *For every nonzero integer c, the point $[c]P$ is finite and has nonzero abscissa. Write*

$$f([c]P) = \frac{1}{x([c]P)} = a_c + b_c\sqrt{2}.$$

Then

$$\min\{v_3(a_c), v_3(b_c)\} = \begin{cases} 0, & c \text{ odd}, \\ 2 + 2v_3(c), & c \text{ even}. \end{cases}$$

In the even case the formal parameter $z = -x/y$ satisfies

$$v(z([c]P)) = 1 + v_3(c).$$

Proof. The displayed equation has good reduction at three: its discriminant is $64D^3$, which is a unit. Direct reduction gives

$$\overline{P} = (\sqrt{2}, 0) \in E(\mathbb{F}_9).$$

This is a nonidentity point of order two, with unit abscissa. Hence every odd multiple of P is finite with unit abscissa.

The ordinate of P has valuation one, since

$$y(P) = \frac{3(-220 + 139\sqrt{2})}{32}$$

and its parenthesized factor has nonzero residue in $\mathbb{F}_9$. The numerator of the tangent slope is a unit: modulo three, $3x(P)^2 - D = -D \neq 0$. Thus

$$\lambda = \frac{3x(P)^2 - D}{2y(P)} \quad \text{satisfies} \quad v(\lambda) = -1.$$

The duplication formula now gives

$$\begin{aligned} v(x([2]P)) &= v(\lambda^2 - 2x(P)) = -2, \\ v(y([2]P)) &= v(\lambda(x(P) - x([2]P)) - y(P)) = -3. \end{aligned}$$

There is no cancellation: the summands in the first difference have valuations $-2, 0$, and those in the second have valuations $-3, 1$. Consequently $v(z([2]P)) = 1$.

For completeness, the needed formal-group valuation is elementary here. The formal group of the integral Weierstrass equation has integral coefficients, and negation sends z to $-z$. Its multiplication series therefore have the form

$$[m](Z) = mZ + Z^3 g_m(Z^2), \qquad g_m(T) \in \mathcal{O}_{K_3}[[T]].$$

If $v(z) \geq 1$ and $3 \nmid m$, the linear term has strictly smaller valuation than every higher term, so $v([m]z) = v(z)$. For $m = 3$, the linear term has valuation $v(z) + 1$, whereas all higher terms have valuation at least $3v(z) > v(z) + 1$. Thus

$$v([3]z) = v(z) + 1.$$

Write $c = 2 \cdot 3^r u$, where $3 \nmid u$. Repeated application to $z([2]P)$ yields $v(z([c]P)) = 1 + r$. In particular, this parameter is nonzero, so $[c]P$ is not the identity. On the formal group, $x = z^{-2}$ times an integral power series with constant term one. It follows that

$$v(f([c]P)) = 2 + 2r.$$

Finally, $1, \sqrt{2}$ form an integral basis for the unramified quadratic extension $K_3/\mathbb{Q}_3$, with linearly independent residues over $\mathbb{F}_3$. For any rational a, b, not both zero, this gives the exact equality

$$v(a + b\sqrt{2}) = \min\{v_3(a), v_3(b)\}.$$

Applying it to the reciprocal abscissas proves both cases. □

Corollary 6.2. *Let $N \geq 1$ and $c \neq 0$. Then*

$$\frac{a_c}{3^N}, \frac{b_c}{3^N} \in \mathbb{Z}_3 \quad \Longleftrightarrow \quad c \text{ is even and } v_3(c) \geq \left\lceil \frac{N}{2} \right\rceil - 1.$$

For a prescribed integer $r \geq 0$, the smallest positive N forcing $v_3(c) \geq r$ is $N = 2r + 1$. At this value the anchor implies

$$|c| \geq 2 \cdot 3^r.$$

Proof. The displayed divisibility condition says that the minimum valuation in Proposition 6.1 is at least N. Since $N > 0$, this excludes the odd case. In the even case it is exactly $2 + 2v_3(c) \geq N$. The stated rounding follows. When $r \geq 1$, the value $N = 2r$ still admits $c = 2 \cdot 3^{r-1}$, so $2r + 1$ is minimal; for $r = 0$, it is the smallest positive value by definition. The final bound follows from the simultaneous divisibility by two and 3^r. □

Application to the physical anchor. In the merged-anchor theorem, replace its arbitrary constant C_* by 3^N and its coefficient $S_0 = C_*^2$ by 3^{2N}. The two-adic argument is unchanged because three is a two-adic unit. Once the root parity and primary divisibility conditions have been extracted, the three-adic argument recovers exactly the two conditions in Corollary 6.2, with $c = a - 1$. The existing point domain requires $C_S = S - P \neq O$, so $c \neq 0$; this condition must be retained. If one instead extended f by $f(O) = 0$, the anchor alone would not exclude $c = 0$.

Suppose the inherited global argument needs

$$|c| > (80H)^{10}, \qquad H < 2^h.$$

It suffices to choose an integer r satisfying

$$2 \cdot 3^r > (80 \cdot 2^h)^{10}$$

and to take

$$N = 2r + 1, \qquad C_* = 3^{2r+1}, \qquad S_0 = 3^{4r+2}.$$

These substitutions change only fixed integer coefficients. They add no coordinate or equation and do not change the polynomial degree. Completeness is retained by requiring the already freely chosen source index c to be divisible by $2 \cdot 3^r$ in the same fixed local congruence conditions. This is a fixed refinement of the construction, not a uniformity assertion with an increasing prime-pattern modulus.

Corollary 6.3. *For $h = 2093448$, the smallest integer r satisfying $2 \cdot 3^r > (80 \cdot 2^h)^{10}$ is*

$$r = 13208226.$$

The corresponding smallest anchor exponent is

$$N = 26416453, \qquad C_* = 3^{26416453}, \qquad S_0 = 3^{52832906}.$$

In particular, a coefficient-height formula $64S_0^2$ specializes to

$$64 \cdot 3^{105665812}.$$

The minimality assertion concerns the stated conservative bound $H < 2^{2093448}$ and this exact anchor criterion.

Proof. The threshold for r is equivalently

$$3^r > 5^{10}2^{20934519}.$$

The following four strict integer comparisons are verified by exact integer exponentiation in the accompanying checker:

$$3^{31867} > 2^{50508}, \quad 3^{15288} > 5^{10}2^{24207},$$
$$3^{15601} < 2^{24727}, \quad 3^{9779} < 5^{10}2^{15477}.$$

The integer identities

$$13208226 = 414 \cdot 31867 + 15288,$$
$$20934519 = 414 \cdot 50508 + 24207$$

give the required lower comparison. Similarly,

$$13208225 = 846 \cdot 15601 + 9779,$$
$$20934519 = 846 \cdot 24727 + 15477$$

give $3^{13208225} < 5^{10}2^{20934519}$ and hence minimality. The remaining values follow from $N = 2r+1$ and $S_0 = 3^{2N}$. □

The exact checker accompanying this calculation verifies the source point, its duplication formulas and valuations, and finite examples of both parity cases using rational arithmetic in $\mathbb{Q}(\sqrt{2})$. The statement for all indices follows from the formal-group proof, not from extrapolating those examples.

7 A scalar normalization without squareclass corrections

Put

$$\beta(X) = 216X^4 + 450459$$

and retain the five-contact polynomial

$$r(T) = T(T-1)(T-2401)(32T+343)(2T-1).$$

Define

$$r_X = r(\beta(X)), \qquad \delta_X = 1 + (17r_X)^2, \qquad R_X = \frac{102^2 r_X}{\delta_X^2}.$$

The scaling factor is an integer. In particular, this normalization introduces no fixed denominator prime.

Lemma 7.1. *For $X \in \mathbb{Q}$ one has*

$$\beta(X) \in \mathbb{Z} \quad \Longleftrightarrow \quad X \in \mathbb{Z}.$$

For every real X, $\beta(X) \geq 450459 > 2401$, so $r_X > 0$ and $\delta_X > 0$. If $p > 76176$ and $v_p(\beta(X)) < 0$, then

$$v_p(\beta(X)) = 4v_p(X), \qquad v_p(r_X) = 5v_p(\beta(X)), \qquad v_p(R_X) = -3v_p(r_X).$$

In particular, all three valuations are even.

Proof. If X is an integer, so is $\beta(X)$. Conversely, write $X = a/b$ in lowest terms with $b > 0$. Integrality of $\beta(X)$ gives $b^4 \mid 216 = 2^3 \cdot 3^3$. Neither prime exponent admits a nontrivial fourth power, so $b = 1$.

The real inequalities follow from the fourth power and the five positive factors of $r(T)$ when $T > 2401$. At a prime in the stated range, a negative valuation of β forces $v_p(X) < 0$. The unique dominant terms in β and $r(\beta)$ are respectively $216X^4$ and $64\beta^5$. This gives the first two valuation formulas. In this case $v_p(r_X) < 0$, so the dominant term of δ_X is $17^2 r_X^2$, and $v_p(\delta_X) = 2v_p(r_X)$. The final formula follows. The displayed multiple of four is even. □

Lemma 7.2. *If $X \in \mathbb{Z}$, then r_X is a nonzero square in $\mathbb{Q}_2$, $\mathbb{Q}_3$, and $\mathbb{Q}_{17}$. More precisely,*

$$v_2(r_X) = 2, \quad r_X/4 \equiv 1 \pmod 8, \qquad v_3(r_X) = 2, \quad r_X/9 \equiv 1 \pmod 3,$$

and at seventeen either $v_{17}(r_X) = 0$ with square-unit residue, or

$$v_{17}(r_X) = 2, \qquad r_X/17^2 \equiv 9 \pmod{17}.$$

Proof. Since 216 is divisible by eight and by twenty-seven,

$$\beta \equiv 3 \pmod 8, \qquad \beta \equiv 18 \pmod{27}.$$

For the two even factors of $r(\beta)$ one has

$$v_2(\beta - 1) = v_2(\beta - 2401) = 1, \qquad \frac{\beta - 1}{2} - \frac{\beta - 2401}{2} = 1200 \equiv 0 \pmod 8.$$

Their odd parts therefore have square product congruent to one modulo eight. The remaining odd factors have product $3 \cdot 7 \cdot 5 \equiv 1 \pmod 8$. This proves the assertions at two. At three, only the factor β is divisible by three; it has valuation two. Reduction after division by nine gives

$$\frac{r(\beta)}{9} \equiv 2 \cdot 2 \cdot 2 \cdot 1 \cdot 2 \equiv 1 \pmod 3.$$

At seventeen the fourth powers are exactly $0, 1, 4, 13, 16$. Since $216 \equiv 12$ and $450459 \equiv 10 \pmod{17}$, all possibilities are given by the table

X^4	0	1	4	13	16
β	10	5	7	13	15
$r(\beta)$	0	15	2	13	2.

The nonzero residues are squares: $13 = 8^2$, $15 = 7^2$, and $2 = 6^2$ modulo seventeen.

In the remaining case $17 \mid X$. The exact identity

$$32 \cdot 450459 + 343 = 17^2 \cdot 49879, \qquad 49879 \equiv 1 \pmod{17},$$

and $v_{17}(X^4) \geq 4$ show that the factor $32\beta + 343 = 6912X^4 + 17^2 \cdot 49879$ has valuation exactly two and unit residue one. The other four factors are units, with product $10 \cdot 9 \cdot 6 \cdot 2 \equiv 9 \pmod{17}$. This proves the last assertion. The usual local square criteria at two and at odd primes now prove all three square statements. □

Proposition 7.3. *For every integer X, the number R_X is prepared: it is positive and less than one, its reduced denominator is a square, it is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and*

$$v_2(R_X) = 4, \qquad v_3(R_X) = 4, \qquad v_{17}(R_X) \in \{2, 4\}.$$

No squareclass correction is required.

Proof. The contact value r_X is a positive integer. One has $\gcd(r_X, \delta_X) = 1$, and the preceding lemma shows that δ_X is a unit at two and three; it is a unit at seventeen by its definition. Thus $\gcd(102^2 r_X, \delta_X^2) = 1$, and the reduced denominator is exactly δ_X^2.

The quotient $R_X / r_X = (102/\delta_X)^2$ is a rational square. Its local valuations add two at each of $2, 3, 17$, proving both the asserted squareclasses and the exact depths. Finally, since $r_X \geq 1$,

$$0 < R_X = \frac{102^2 r_X}{(1 + 17^2 r_X^2)^2} < \frac{102^2}{17^4 r_X^3} \leq \frac{10404}{83521} < 1.$$

□

7.1 The norm tree and its domain

Fix a positive rational number ρ and six rational tests $t_0, \ldots, t_5$, with their assigned positive integer orders. Start the scalar norm tree with the distinguished node $1 + t_0^2/\rho$. Its other initial nodes are the assigned normalized tests or pair norms. Every later operation is

$$N_\rho(u, v) = u^2 + \rho v^2$$

or the unchanged carrying of an unpaired node. This is the constant-left tree used by the scalar certificate; denote its root by W_ρ.

Lemma 7.4. *Where its rational expressions are defined, $W_\rho > 0$. When all six tests tend to zero, W_ρ tends to one. For fixed nonzero rational ρ, at any finite set of primes the tests can be required sufficiently small that W_ρ is a unit at every prime in that set.*

Proof. The distinguished initial node is positive. At each norm operation its positive descendant contributes a positive square, with a positive coefficient if it is the second argument. Its path therefore remains positive up to the root. When all tests vanish, this path starts at one and all other nodes vanish, so its root is one. For fixed ρ, the tree is a polynomial in the normalized tests with fixed rational coefficients. It is continuous at zero over $\mathbb{R}$ and over every $\mathbb{Q}_p$. Its value at zero is one. The inverse image of $\mathbb{Z}_p^*$ is therefore a neighborhood of zero, proving the final claim. □

There are two valid implementations. One may replace the contact parameter throughout the tree by $\rho = R_X$ and put $b = R_X W_{R_X}^2$. Alternatively, one may retain the original tree with $\rho = r_X$ and normalize only its final pole input:

$$b = R_X W_{r_X}^2.$$

The second choice retains the original test denominators. It also has the quadratic scaling equations

$$\delta_X = 1 + 17^2 r_X^2, \qquad \delta_X V = 102 W_{r_X}, \qquad b = r_X V^2,$$

where the last expression can use the existing square-and-product gates. The first two displayed equations are quadratic. The soundness and completeness proof for this substitution is given below.

For rational inputs, both implementations have $r_X, R_X, \delta_X > 0$ and $W_\rho > 0$. Thus the scalar contact, normalizer, norm-root, and pole-input nonvanishing conditions follow on the stated rational-expression domain. A coordinate used solely to certify one of these nonzero conditions can be removed after its occurrence condition is checked. This observation does not remove unrelated denominator guards in the elliptic-point formulas.

7.2 Soundness and simultaneous completeness

Proposition 7.5. *At every finite scalar contact prime $p > 76176$, the replacement has the same positive odd contact valuation as the original certificate. Consequently it preserves its norm-tree extraction of the assigned contact orders, for either implementation above.*

Proof. At such a contact, $\nu = v_p(r_X) > 0$ is odd. The normalizer satisfies $\delta_X \equiv 1 \pmod p$, and 102 is a unit. Hence

$$v_p(R_X) = v_p(r_X) = \nu, \qquad R_X / r_X \in \mathbb{Z}_p^{*2}.$$

For either $\rho = r_X$ or R_X, the two terms in $u^2 + \rho v^2$ have valuations of opposite parity when both are nonzero. There is therefore no cancellation: its valuation is the minimum of $2v_p(u)$ and $\nu + 2v_p(v)$. Every step of the existing constant-left extraction uses exactly this identity, the assigned powers of the contact parameter in the leaves, and the odd-pole exclusion for b. All these valuations are unchanged. Lemma 7.1 also shows that a contact with infinity has even valuation outside the fixed exceptional set, so it creates no omitted odd contact. The inherited height and comparison arguments therefore give $\beta \in \mathbb{Z}$, and Lemma 7.1 gives $X \in \mathbb{Z}$. □

Proposition 7.6. *Under the same arithmetic inputs as the source construction, the scalar block with the input map and normalization above can be satisfied at every integer input, together with the remaining point, height, comparison, and size conditions. The finite squareclass correction parameter can be omitted.*

Proof. Fix an integer input. The numbers r_X, R_X, δ_X are now fixed, before any contact primes are chosen. Proposition 7.3 supplies the sign, local squareclasses, square denominator, and small positive real size formerly obtained by a correction. Include the finite support of the numerator and denominator of R_X, and that of ρ, in the fixed local approximation conditions for the common point H and the scalar point $S - P$.

The required approximation is the same one used in the source. Choose H sufficiently deep at these finite places and in a small real neighborhood of the identity, excluding the identity itself. Shrink the real neighborhood so that its formal parameter is bounded away from zero. Choose $S - P$ still deeper at every fixed finite place and sufficiently close to the real identity. The reciprocal-abscissa tests at these points then tend to zero, and the quotient of their formal parameters tends to zero with H in that smaller neighborhood. Fixed powers of ρ in the denominators do not affect this convergence.

One can impose these conditions with the required congruences on the integer indices: the source first chooses the fixed modulus and depths, then applies its prime-pattern and common-point construction. Increasing the modulus to include the finite support of δ_X is allowed because this support was fixed before that application. It introduces no new family of varying-prime conditions and no stronger prime-pattern theorem. The real compact-group argument permits the same shrinking of the neighborhoods and arbitrarily large scalar index. All other source index and comparison choices are unchanged.

By Lemma 7.4, these choices make W_ρ a unit at every prime in the support of R_X and give $|W_\rho - 1| < 1/2$ at the real place. Therefore

$$0 < b = R_X W_\rho^2 < 9/4 < 17.$$

The factor W_ρ^2 preserves the local squareclasses and, at $2, 3, 17$, preserves the depths of Proposition 7.3. The reduced denominator of b is a square: at a prime in the support of R_X, the root is a unit, while at every other prime its valuation contribution is even. Thus b is prepared. The same inherited rational pole-realization theorem supplies the final pole witnesses. This proves completeness without a correction branch. □

The input polynomial has degree four and can be evaluated with two quadratic gates while preserving the source expression $\beta = 2Y + 3$:

$$Z = X^2, \qquad Y = 108Z^2 + 225228.$$

The gain in any compiled polynomial also depends on its exact occurrence and denominator inventory; the preceding propositions establish the arithmetic substitution independently of that count.

7.3 Number-field scope

All displayed algebraic identities remain valid in characteristic zero. Positivity of the contact and norm root also holds in a number field with a real embedding, after applying that embedding. The local squareclass table, however, concerns rational integer inputs and the three specified completions of $\mathbb{Q}$. It does not establish the corresponding preparation assertions for every algebraic integer in a larger field.

Even the integrality equivalence of Lemma 7.1 does not automatically become an equivalence for rings of integers. Let x_0 be a real fourth root of $1/108$ and $L = \mathbb{Q}(x_0)$. Then

$$\beta(x_0) = 450461 \in \mathbb{Z}.$$

But x_0 is not an algebraic integer, since $x_0^4 = 1/108$. Thus this elementary input map alone does not transfer a definition to $\mathcal{O}_L$. No assertion that the complete rational defining polynomial defines every $\mathcal{O}_L$ is made or follows from this normalization.

8 Asymmetric formal depth for the quartic

The finite algebra constructions use truncated logarithms on both families of elliptic points. The quartic can avoid the five primary logarithms by imposing a larger formal depth. We give the local argument and identify exactly the extra prime-selection input. These are the asymmetric-depth statements of [2, Section "Replacing five logarithms by asymmetric formal depth"].

Lemma 8.1. *Under the group hypothesis $2E(K) = \mathbb{Z}P$, a finite nonidentity point of $E(K)$ with nonzero square abscissa belongs to $\mathbb{Z}P$. Conversely every nonzero multiple of P has a representation*

$$x = h^2, \qquad j^2 = x^2 - D, \qquad y = hj$$

with $h, j \in K^\times$.

Proof. Since $P = 2R_0$ and P has infinite order, the group hypothesis gives $E(K) = \mathbb{Z}R_0 \oplus E[2](K)$, with four rational two-torsion points. The abscissa map for the two-isogeny is the homomorphism to $K^\times/K^{\times 2}$ that takes Q to $[x(Q)]$, takes $(0,0)$ to $[-D] = [-1]$, and takes the identity to $[1]$; see [5, Chapter X, Section 4]. It kills $2E(K)$. Its values at R_0, $(0,0)$, and $(d,0)$ are $[1+\sqrt{2}]$, $[-1]$, and $[d]$. The first two classes are independent by their signs at the two real embeddings. The third has odd valuation at the prime (d) above seven and is independent of both. Thus the image has eight elements, equal to $|E(K)/2E(K)|$, and its kernel is exactly $2E(K)$.

The duplication formula makes the abscissa of every nonzero double a square. Choose its nonzero square root h and put $j = y/h$. Its curve equation gives $j^2 = x^2 - D$. Neither h nor j vanishes on a nonzero cyclic multiple, since such a point is not two-torsion. □

The logarithm truncation. For the curve $y^2 = x^3 - Dx$, put $f(Q) = 1/x(Q)$ and $z(Q) = -x(Q)/y(Q)$. Define

$$G_9(W) = \sum_{j=0}^{9} a_j W^j, \qquad a_0 = 1, \quad a_1 = -\frac{2}{5},$$

$$a_j = \frac{4j-5}{4j+1} a_{j-1} \quad (2 \le j \le 9), \qquad \ell_{40}(Q) = z(Q) G_9\big(Df(Q)^2\big).$$

These definitions specify all coefficients used by the five alternate logarithm circuits.

To verify the formula, let $A(W) = \sum_{j \ge 0} a_j W^j$, with the same recurrence continued for every $j \ge 2$. The curve equation gives $z^2 = f/(1 - Df^2)$, while the invariant differential satisfies $dx/(2y) = z\,df/(2f)$. Substitution shows that $d(zA(Df^2)) = dx/(2y)$ precisely when

$$4W(1-W)A'(W) + (1+W)A(W) = 1 - W.$$

The coefficient of W^0 gives $a_0 = 1$, the coefficient of W gives $a_1 = -2/5$, and every subsequent coefficient gives the displayed recurrence. The unique formal logarithm with linear coefficient one is therefore $zA(Df^2)$.

At an unramified place above $p > 76176$ where D is a unit and $v_p(z(Q)) = r \ge 1$, one has $v_p(Df(Q)^2) = 4r$. The recurrence gives

$$v_p(a_j) \ge -\frac{4j+1}{p-1},$$

and a_j is p-integral if $4j + 1 < p$. Indeed its denominator divides a product of integers $4k+1$ for $1 \le k \le j$, and the sum of the numbers of their multiples of $p, p^2, \ldots$ is at most $(4j+1)/(p-1)$. These inequalities show that the terms with $j \ge 10$ have valuation at least $40r$. Thus the truncation has relative precision at least forty, as used in the comparison argument.

Write $U_0 = H = [h]P$, $U_i = [u_i]P$, $V_0 = H - P$, and $V_i = U_i - Q_i$, where $Q_i = [n_i]P$. Put

$$L_i = z(U_i)/z(H), \qquad L_i' = \sigma\big(\ell_{40}(V_i)/\ell_{40}(V_0)\big).$$

The nine order-one tests are $f(U_i)$ for $1 \le i \le 4$ and $\sigma f(V_i)$ for $0 \le i \le 4$. The seven order-forty tests are $f(H)$, the four $L_i - L_i'$, $L_1 - \beta$, and the direct Lucas residual. The scalar ratio retains order 31.

Lemma 8.2. *At a consumed unramified prime $p > 76176$, these tests imply $L_i \equiv n_i \pmod{p^{40}}$ at the selected place.*

Proof. For a formal point with $r = v(z(Q)) \geq 1$, the formal logarithm of this curve satisfies

$$\log_E(Q)/z(Q) \in 1 + p^{4r}\mathcal{O}.$$

Indeed its first correction is of order four in z, and the coefficient bound $v(a_j) \geq -(4j+1)/(p-1)$ ensures that later terms have at least that valuation; the coefficients with $4j+1 < p$ are integral. The large-prime bound makes the same inequality hold for all the remaining terms. The omitted tail in ℓ_{40} similarly has relative precision forty.

The test $v(f(H)) \geq 40$ gives $v(z(H)) \geq 20$. The full logarithm ratios are u_i/h and $(u_i - n_i)/(h-1)$. At least one of $h, h-1$ is a p-unit. The corresponding ratio is integral, and the relative-unit estimates show that its truncated ratio has the same valuation. The cross-test transfers integrality to the other ratio. Consequently $v(z(U_i)) \geq v(z(H)) \geq 20$ as well. The primary relative error is now divisible by p^{80}, and the alternate error by p^{40}. Since the ratios are integral, their absolute errors have the same bounds. Multiplying the two ratio congruences by h and $h-1$ and subtracting yields the assertion. At an inert prime conjugation preserves the valuation, so the identical argument applies. □

For completeness we use the following explicit weighted form of the prime-pattern input. For distinct fixed rational coefficients c_j, after the prescribed congruences and obstruction removal, the patterns $p_j = u + c_j w^2$ with $u \in [N/2, N]$ and $1 \leq w \leq \tau(N)\sqrt{N}$ have total weight at least $c_0 N^{3/2}\tau(N)$ for a fixed $c_0 > 0$. Their individual weights are $O((\log N)^k)$ for k forms, and $\tau(N)$ can tend to zero arbitrarily slowly in the diagonal choice. This is the source's fixed-field weighted count; it is an arithmetic input here.

Lemma 8.3. *Under this weighted input the selected primes may be chosen so that no selected prime divides any opposite reduced group order.*

Proof. Require additionally $\tau(N)N^{1/6}/(\log N)^k \longrightarrow \infty$. The strip $w < N^{1/3}$ has at most $O(N^{4/3})$ pairs and weighted size $O(N^{4/3}(\log N)^k)$, negligible compared with the retained weight of order $N\tau(N)\sqrt{N}$. An admissible pattern therefore remains with $N^{2/3} \leq w^2 = o(N)$.

All selected primes are asymptotic to u. By Hasse's bound [5, Theorem V.1.1], every opposite group order is $d_+(p_j) = p_j + O(\sqrt{N})$. For large N the ratio $d_+(p_j)/p_i$ lies strictly between $1/2$ and 2, so divisibility would force equality. For $i \neq j$, equality contradicts $|p_i - p_j| \geq c_* N^{2/3}$, where $c_* = \min_{i \neq j} |c_i - c_j| > 0$. For $i = j$ it is impossible because $4 \mid d_+(p_i)$ and p_i is odd. The one-form case needs only this last argument. □

Let $M_\pm = \operatorname{lcm}_p(d_\pm(p)/4)$ and $\Pi = \prod_p p$. The original order and avoidance conditions give $\gcd(M_-, M_+) = 1$, $\gcd(M_-, L) = 16$, and $\gcd(M_+, L) = 1$ for the previously fixed modulus L. The lemma gives $\gcd(\Pi, M_+) = 1$, and the primes may be chosen beyond the factors of L. Thus the congruences

$$h \equiv 0 \pmod{\Pi^{19} M_-}, \qquad h \equiv 1 \pmod{M_+}, \qquad h \equiv h_* \pmod{L}$$

are compatible when $64 \mid h_*$. At each primary place the factor M_- first makes the point formal, and the additional p^{19} raises its depth from at least one to at least twenty by the formal logarithm isometry [5, Theorem IV.6.4]. The real approximation and the fixed scalar conditions are preserved on this progression. This gives the required primary depth while retaining the opposite depth and all completeness conditions. The 29- and 31-variable constructions do not need this weighted strengthening.

8.1 The height inequalities for the quartic

For the quartic use the fixed anchor

$$C_* = 3^{26416453}, \qquad S_0 = C_*^2 = 3^{52832906}.$$

The exact three-adic index calculation and integer comparisons of Section 6 give

$$c = |a-1| \geq 2 \cdot 3^{13208226} > (80H)^{10}, \qquad 1 \leq H < 2^{2093448}.$$

The scalar order 31 gives $h(\beta) \leq Hc^{19/31}$. Consequently

$$|a| \geq c/2 > (40H)^{10}, \qquad h(\beta) < c/80 \leq |a|/40, \quad \log(40H) < c/80 \leq |a|/40.$$

Indeed $c^{12/31} > (80H)^{120/31} > 80H$ proves the scalar inequality. Also $c/80 > (80H)^{10}/80 > 40H > \log(40H)$.

Put $B = \max(|n|, |a|, |t|, |b|)$ and let F be the Lucas residual. If $F \neq 0$, then $|F| < 5B^4$ and the order-forty tests, the product-height identity and contact height give

$$B^2 - 6 < 7H(5B^4)^{19/40}, \qquad B^{1/10} < 7H5^{19/40} + 6 < 24H.$$

This contradicts the anchor. Thus $F = 0$, and Lemma 2.1 gives $\log B > |a| - 3$. If $\beta \neq n$, use the weaker order-twelve consequence of the order-forty input tests. The reduced numerator of $\beta - n$ is at most $2e^{h(\beta)}B$, so

$$B^{5/12} < 40He^{19h(\beta)/12}, \qquad \log B < \frac{12}{5}\log(40H) + \frac{19}{5}h(\beta) < \frac{31}{200}|a| < |a| - 3.$$

This is the second contradiction. The argument forces $\beta \in \mathbb{Z}$ independently of the particular input map, provided its scalar contacts and integer-input witnesses obey the stated conditions. Section 7 proves these conditions for the chosen fourth-power map.

9 Quadratic circuits and the explicit quartic

A quadratic circuit is a finite list of polynomial equations of total degree at most two, with a designated input coordinate. Arithmetic assignments and division equations are used only with their stated nonzero domains. Each of the algebraic transformations below preserves the rational input projection on its stated nonzero domain. The arithmetic identification of the upstream projection with $\mathbb{Z}$ is a separate input.

9.1 Direct primary ratios

Lemma 9.1. *Let h_0, j_0, h_i, j_i be nonzero elements of a field, for $1 \leq i \leq r$. The equations*

$$z_i j_i = -h_i \quad (0 \leq i \leq r), \qquad L_i z_0 = z_i \quad (1 \leq i \leq r)$$

may be replaced by

$$A_i = j_i h_0, \qquad A_i L_i = h_i j_0 \quad (1 \leq i \leq r).$$

The two systems have identical projections onto the h, j, L coordinates, and every extension in either system is unique. They have respectively $2r+1$ and $2r$ auxiliary field coordinates, counting the L_i in both cases.

Proof. The first system uniquely determines $z_i = -h_i/j_i$ and $L_i = h_i j_0/(j_i h_0)$. In the second, the uniquely determined $A_i = j_i h_0$ is nonzero, and the same formula for L_i follows. All displayed equations are quadratic. □

Apply the lemma over $K = \mathbb{Q}(\sqrt{2})$ to the five primary points. In their square-abscissa charts, $x_i = h_i^2$, $j_i^2 = x_i^2 - D$, $y_i = h_i j_i$, and $z_i = -h_i/j_i$. The retained source domain makes every h_i, j_i nonzero. The intermediate z_i occur only in these four ratios; the scalar block uses its own direct expression for the needed ratio. Replacing five z_i and four L_i by four A_i and four L_i therefore saves one K-coordinate, hence two rational coordinates. The lemma supplies both reconstruction maps, including the nonzero conditions needed for their validity.

9.2 Evaluating products in smaller blocks

For a list $y_1, \dots, y_n \in K$ define

$$F(q) = \prod_{i=1}^{n} (1 + y_i q).$$

The y_i in the detectors are squares of the field-valued tests. For a partition $I_1 \sqcup \cdots \sqcup I_s = \{1, \dots, n\}$, put $F_j(q) = \prod_{i \in I_j} (1 + y_i q)$. Then

$$F(q) = \prod_{j=1}^{s} F_j(q) \tag{1}$$

is an identity in the polynomial ring $\mathbb{Z}[y_1, \dots, y_n, q]$.

The source has nine tests of order one and seven of order forty. Their coefficients are shared among three detector evaluations. For the nine tests, use three consecutive blocks of size three. For the seven tests, use blocks of sizes three and four. Compute the coefficients of each block once by multiplication of its linear factors. Within each detector compute the necessary powers of its rational weight, evaluate the block polynomials, and multiply their values. Equation (1) shows that the resulting W is exactly unchanged, for every assignment of the test values and weights. No local or height estimate is altered by this operation.

Every multiplication in K is written in the fixed basis:

$$(a_0 + a_1\sqrt{2})(b_0 + b_1\sqrt{2}) = (a_0 b_0 + 2a_1 b_1) + (a_0 b_1 + a_1 b_0)\sqrt{2}.$$

Each assigned component consequently has a quadratic defining equation. The following count includes only the shared coefficient calculations and their three evaluations; the test squares and final product between the two order groups are unchanged.

number of factors	full coefficients	smaller blocks	difference
9	66	54	12
7	49	43	6

For example, three cubic blocks require nine field multiplications for their shared coefficients, hence eighteen rational coordinates. Each evaluation requires two rational powers, six component outputs, and four component coordinates for the two products: twelve rational coordinates. The total is $18 + 3 \cdot 12 = 54$. The analogous $3 + 4$ partition requires sixteen shared coordinates and nine coordinates per evaluation, giving $16 + 3 \cdot 9 = 43$.

These elementary circuit counts precede the final rational changes of coordinates. Four final simplifications available in the full coefficient presentation use coefficients absent from the smaller blocks. The net saving in the finished system is therefore fourteen, rather than eighteen. The exact certificates include the five remaining simplifications, so this count does not rely on an estimate of compiler behavior.

9.3 Reusing fourth powers in the cubic blocks

For three inputs y_1, y_2, y_3, write

$$(1 + y_1 q)(1 + y_2 q)(1 + y_3 q) = 1 + e_1 q + e_2 q^2 + e_3 q^3.$$

Their coefficients may be computed from

$$e_1 = y_1 + y_2 + y_3, \qquad e_2 = y_1 y_2 + (y_1 + y_2) y_3, \qquad e_3 = e_2 y_3 - (y_1 + y_2) y_3^2.$$

If y_3^2 is already a coordinate, each of the last two identities is a quadratic defining equation for one new field-valued coordinate. The usual coefficient circuit keeps $y_1 y_2$ as a separate

intermediate in addition to e_2, e_3. The two displayed equations need no such intermediate. The last identity follows by substituting the expression for e_2 and canceling its second summand.

In the nine order-one tests, each of three cubic blocks can place an alternate reciprocal-abscissa square in the third position. The five alternate logarithm circuits already compute

$$w^2 = (Df(V)^2)^2 = D^2 f(V)^4.$$

Consequently the required $y_3^2 = \sigma(f(V)^4)$ is an affine expression in an existing field-valued coordinate:

$$y_3^2 = \sigma\left(\frac{w^2}{D^2}\right), \qquad \frac{1}{D^2} = \frac{193 + 132\sqrt{2}}{2401}.$$

Division by this fixed nonzero coefficient and conjugation are rational linear operations. Their order in this formula is specified: the fixed inverse is multiplied before conjugation. Each cubic block therefore saves one K-coordinate, or two rational coordinates. Three such blocks save six coordinates.

9.4 Constant pivots and the final polynomial

Every constant-pivot elimination used in the coefficient certificate starts from a retained equation $aZ + F = 0$, $a \in \mathbb{Q}^\times$. Substitute its unique value $Z = -F/a$ into the other equations. An equation may also be changed by polynomial multiples of other retained equations. These operations preserve the common zero set, with unique reconstruction of Z. Their particular uses are checked to leave all remaining equations quadratic. Fixed nonzero rational rescaling and primitive integer normalization of an equation likewise preserve its zeros.

The parity and primary-divisibility restriction is treated separately in Theorem 5.3: it preserves soundness and has witnesses at every integer input, rather than preserving every unrestricted auxiliary solution. The scalar input and normalization have the analogous separately proved completeness property. Combining these arithmetic steps with the identities of this section gives the exact quartic of Theorem 11.4.

10 The input circuit and fixed coefficients

We now combine the preceding parity and circuit changes with an input map whose scalar contact is already a square in the three required local fields. There is then no finite correction parameter. The normalization in this section uses only an integer scaling factor and introduces no fixed denominator prime.

The input map and the normalized contact are

$$\beta = 216X^4 + 450459, \qquad r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1), \qquad R = \frac{102^2 r}{(1 + (17r)^2)^2}.$$

Their integrality, local-square and denominator assertions were proved in Section 7.

Soundness and completeness. Use R as the contact parameter throughout the existing scalar norm tree, including its assigned denominator powers. Write its root as W_R and its pole input as $b = RW_R^2$. For rational inputs $R > 0$. The tree starts with a nonzero constant leaf, and each norm operation is $u^2 + Rv^2$; thus $W_R > 0$ on its stated rational-expression domain. In particular, $b \neq 0$ requires no additional inverse coordinate.

At every consumed scalar contact, Proposition 7.5 preserves the positive odd contact valuation. The two terms of $u^2 + Rv^2$ then have opposite valuation parity, and the unchanged minimum-valuation argument extracts the same assigned contact orders. The source height

and comparison estimates are written in terms of $\mathrm{h}(\beta)$ itself, so their numerical constants are unchanged. They imply $\beta \in \mathbb{Z}$; Lemma 7.1 then implies $X \in \mathbb{Z}$.

For completeness, fix an integral input first. The rational number R and its numerator and denominator support are now fixed before the prime-pattern construction is applied. The same simultaneous approximation used in the source can make the six scalar tests sufficiently small that W_R is a unit at every prime in that support and $|W_R - 1| < 1/2$ at the real place. The permitted fixed modulus is enlarged by this finite support. No new condition on the later varying contact primes is needed. It follows that

$$0 < b = RW_R^2 < 9/4 < 17.$$

Its local squareclasses, depths at $2, 3, 17$, and square reduced denominator have exactly the required properties. The inherited prepared pole-realization theorem therefore applies with preparation parameter zero. The common primary index can still be chosen divisible by 64, as required in Section 5. This is a direct application of the original soundness and completeness arguments to the new scalar parameter. It does not claim that arbitrary old auxiliary solutions are preserved.

Quadratic circuit and variable count. In the 474-coordinate source, let Y denote its former shifted input-square coordinate, so all downstream occurrences of the input parameter are $2Y + 3$. Add only one coordinate Z and use

$$Z = X^2, \qquad Y = 108Z^2 + 225228.$$

Thus $2Y + 3 = 216X^4 + 450459$. Set all seven remaining correction coordinates to one and delete their seven equations. The corrected contact gate then has a constant-linear defining equation; substitute its unique affine value and delete that coordinate and equation. These steps save eight coordinates and add one.

After an invertible affine change in the existing raw-contact coordinate, the normalization uses two added coordinates, for δ and R, with the three quadratic equations

$$R\iota = 1, \qquad \delta - 1 - 289r^2 = 0, \qquad 10404r\iota - \delta^2 = 0.$$

Here ι is the already existing inverse coordinate, now interpreted as $1/R$. The first equation is its retained defining equation. The other two equations introduce the normalization without a variable denominator. Preparation zero deletes the old four preparation coordinates and three equations; substitution of $U + b = 2b$ in every remaining occurrence gives $U = b$.

The total count is therefore

$$474 - 8 + 1 - 4 + 2 = 465.$$

There are 435 quadratic residuals. This is the quadratic system before the final power and pole simplifications below.

The smaller fixed anchor. The parity and primary-index arguments hold for any fixed $C_* = 3^N$ and $S = C_*^2$. The exact local depth result of Corollary 6.2 shows that the physical anchor, with the retained condition $c = a - 1 \neq 0$, forces

$$|c| \geq 2 \cdot 3^r \qquad \text{when} \qquad N = 2r + 1.$$

The retained height constant is exactly

$$\mathcal{H} = 2^{110}210^{1800}\left(\prod_{\substack{p \leq 76176 \\ p \text{ prime}}} p\right)^{19} + 128.$$

An exhaustive prime enumeration and exact integer calculation give 7499 primes in the product and show that $\mathcal{H}$ has 2093448 binary digits. Thus $\mathcal{H} < 2^{2093448}$. Using this sharper bound, Corollary 6.3 gives the certified choice

$$r = 13208226, \qquad N = 26416453, \qquad S = 3^{52832906}.$$

It yields $|c| > (80\mathcal{H})^{10}$ and hence also $|c|^{12/31} > 80\mathcal{H}$, as required by the inherited height contradictions. Completeness still allows the scalar index to be chosen with this fixed divisibility. No coordinate, residual, or formal coefficient in $\mathbb{Z}[S]$ changes. Only the fixed integer substituted for S is reduced.

11 A quartic in 456 total variables

We give two further reductions of the quadratic system: a more economical inverse-power calculation and a rescaling of the four pole certificates. The rescaling is an exact equivalence on the already enforced nonzero domain. The change of scalar order is justified by the local extraction argument and the same global height bounds. Neither reduction introduces an additional arithmetic hypothesis.

11.1 Inverse powers with quadratic division equations

Lemma 11.1. *Let F be a field and $R \in F^\times$. Starting with $v_0 = R^{-1}$, the r quadratic equations*

$$Rv_j = v_{j-1}^2 \qquad (1 \le j \le r)$$

have a unique solution. It is

$$v_j = R^{-(2^{j+1}-1)}.$$

In particular, R^{-31} requires four additional coordinates when R^{-1} is already retained.

Proof. Every equation has the nonzero coefficient R at its new coordinate, so it determines that coordinate uniquely. If $v_{j-1} = R^{-a}$, then $v_j = R^{-(2a+1)}$. Starting from $a = 1$ gives $1, 3, 7, 15, 31, \ldots$, which proves the formula by induction. □

The scalar order may therefore be reduced from 32 to 31, replacing five ordinary squaring coordinates by four division coordinates. The two normalized ratio components become D/R^{31}; their weighted squares have the exact form

$$R(D/R^{31})^2 = D^2/R^{61}.$$

The odd exponent 61 retains the valuation-parity property of the scalar certificate. More generally, this argument works with every positive assigned order, as the next lemma makes explicit.

Lemma 11.2. *Let p be a prime, let $R \in \mathbb{Q}^\times$ have positive odd valuation $\nu = v_p(R)$, and form a finite binary expression by applying $(a, b) \mapsto a^2 + Rb^2$ to rational leaves. Suppose a leaf is $L = D/R^m$, with $m \ge 1$, and let W be the root. If*

$$v_p(RW^2) \ge 0,$$

then

$$v_p(D) \ge (m-1)\nu + 1 \ge m.$$

Proof. At every internal node the two summands have valuations of opposite parity, so they do not cancel. A leaf at depth j contributes a candidate valuation

$$2^j v_p(L) + e\nu, \qquad 0 \le e < 2^j.$$

The root valuation is the minimum of these candidates, by induction up the tree. Since ν is odd, the assumed inequality gives $v_p(W) > -\nu/2$. Therefore every candidate exceeds $-\nu/2$, and in particular

$$v_p(L) > -\frac{e + 1/2}{2^j}\nu > -\nu.$$

Substituting $v_p(L) = v_p(D) - m\nu$ and using the integrality of valuations gives the result. This proof uses no parity condition on m. □

For the scalar point indices, this gives $M_5(\beta)^{31} \mid c$, where $c = a - 1 \neq 0$. Under the retained contact-height input, it follows that

$$h(\beta) \le H|c|^{19/31}.$$

The existing anchor gives $|c| > (80H)^{10}$, with $H \ge 1$. Hence

$$|c|^{12/31} > (80H)^{120/31} > 80H, \qquad h(\beta) < |c|/80.$$

The order-forty comparison tests, the product-height identity, and the two Lucas height contradictions are unchanged. This is the same scalar upper bound used there. For integer-input completeness, R is fixed before the scalar approximation conditions are chosen. The normalized leaves can be made sufficiently small at the real places and sufficiently deep at the finitely many fixed primes for order 31, just as for any fixed positive order. Thus this reduction preserves both arithmetic implications under exactly the previous inputs.

11.2 A smaller quadratic model for the pole certificate

It is important to distinguish the retained scaled conic coordinate from the dimensionless parameter sometimes denoted by e. We write the retained coordinate as E. For a nonzero pole input b, the two equations before introducing the square and product coordinates are

$$\begin{aligned} 17bp^2 &= (17b - E)(17b + 3E), \\ 17bq^2 &= (s^2 - H_0)(s^2 + 3H_0), \qquad H_0 = E(17b - b^2). \end{aligned} \tag{2}$$

These are the prepared equations with preparation parameter zero.

Proposition 11.3. *Over any field of characteristic zero, for every $b \neq 0$, the solutions of (2) correspond bijectively to the solutions in seven coordinates u, p, v, w, h, z, F of the five quadratic equations*

$$\begin{aligned} h &= u(17 - b), \\ z &= v^2, \\ F &= (z - h)(z + 3h), \\ 17p^2 - 289b + (3u - 34)(17u - h) &= 0, \\ 17w^2 - bF &= 0. \end{aligned} \tag{3}$$

The correspondence on the original conic coordinates is

$$E = bu, \qquad s = bv, \qquad q = bw.$$

If b is an affine expression in the remaining coordinates, all five equations have total degree at most two in the full coordinate list.

Proof. The first rescaling gives $H_0 = b^2u(17-b) = b^2h$. Dividing the first equation of (2) by b, and the second by b^3, gives

$$17p^2 = b(17-u)(17+3u), \qquad 17w^2 = b(v^2-h)(v^2+3h).$$

The equations defining z, F express the latter relation quadratically. For the former, put

$$C = 17p^2 - b(289+34u-3u^2), \qquad G = h - 17u + bu.$$

Then the exact polynomial identity

$$C - (3u-34)G = 17p^2 - 289b + (3u-34)(17u-h)$$

removes the cubic term, using the already retained equation $G = 0$. This proves the equivalence of the displayed systems. The maps $u = E/b$, $v = s/b$, $w = q/b$ and their inverses are rational and mutually inverse, while h, z, F are uniquely determined. The degree assertion follows directly from (3). □

The preceding realization uses four base conic coordinates and only three auxiliary coordinates. The former quadratic realization used the same four base coordinates and five auxiliary coordinates for p^2, b^2, H_0, s^2, q^2 (up to fixed nonzero rational rescalings). It therefore saves two coordinates and two equations per pole block. There are four blocks, so the saving is eight coordinates.

The nonzero hypothesis is already enforced in every block. In a comparison block write $W = w_0 + w_1\sqrt{2}$. The retained inverse of w_0 makes $w_0 \neq 0$. Consequently $w_0^2 - 2w_1^2 \neq 0$ over $\mathbb{Q}$. The equation $2w_0T = w_0^2 - 2w_1^2$ then gives $T \neq 0$, and the existing equation expressing b as T^2 divided by the weight and its contact difference forces $b \neq 0$. This is an algebraic argument, independent of the global height hypotheses.

For the scalar block, the fourth-power input map makes its contact r positive, and the normalization

$$R = \frac{102^2 r}{(1+(17r)^2)^2}$$

makes $R > 0$ for every rational input. The scalar binary expression starts with the leaf 1, and each operation is $a^2 + Rb^2$. Its root W is therefore positive. Thus its pole input $b = RW^2$ is positive, in particular nonzero. The rescaling loses no scalar solution either.

11.3 The resulting coefficient table

The construction is obtained directly from the valid 474-coordinate source. The fourth-power input, removal of the scalar correction coordinates, zero preparation, and integral scalar normalization give $474 - 8 + 1 - 4 + 2 = 465$ coordinates. The inverse-power calculation saves one more, and the four pole rescalings save eight. The final count is

$$474 - 8 + 1 - 4 + 2 - 1 - 8 = 456.$$

There are 426 quadratic residual equations. Each is cleared of its fixed rational denominators and divided by the greatest common divisor of its integer coefficients. No parameter-dependent division is used in this primitive normalization.

Theorem 11.4. *Let $R_1, \dots, R_{426}$ be the explicitly supplied primitive quadratic residuals, in 456 total rational coordinates including the input X, and set*

$$Q_{456} = \sum_{i=1}^{426} R_i^2.$$

Its zero set over $\mathbb{Q}$ is exactly the zero set of the displayed quadratic system. It has degree exactly four, content one, 27,784 ordinary monomials, and coefficient height

$$H(Q_{456}) = 64 \cdot 3^{105665812}.$$

The fixed integer coefficient parameter is $S = 3^{52832906}$. Under the four remaining stated source arithmetic inputs and the weighted prime-selection input of the quartic construction, together with the proved Lucas estimate, the existential input projection of this polynomial is $\mathbb{Z}$.

Proof. The equivalence of the single equation and the quadratic system follows from the ordering of $\mathbb{Q}$. The preceding lemmas prove the new algebraic transformations and the order-31 arithmetic implications. All other point, input, local, height, and completeness arguments of the construction remain unchanged.

The complete coefficient list is obtained by squaring the 426 residuals and collecting terms. Every residual has degree at most two, and the coefficient of X^4 in the sum is one. Thus the degree is four and the content is one. The table has 27,791 nonzero terms over $\mathbb{Z}[S]$, which collect into 27,784 ordinary coordinate monomials. The coefficient of the square of the first translated anchor coordinate is exactly $64S^2$. Every other coefficient is $c_0 + c_1 S + c_2 S^2$, where

$$|c_2| \le 8, \qquad |c_0| + |c_1| < 2^{767} < S.$$

Its absolute value is less than $9S^2$. These bounds also show that none of the recorded nonzero coefficients becomes zero at the fixed integer S. This proves the height and monomial assertions.

The independent checker imports no producer code. It verifies every unchanged source row in named coordinates, the four full rescaling identities including the cubic cancellation, all inverse-power equations and their exponents, primitive normalization, and every coefficient of the quartic. These finite checks certify the stated polynomial data. The arithmetic projection assertion retains precisely the hypotheses displayed in its statement. □

12 The complete coefficient table

The computation directory supplies all quadratic residuals and every coefficient of their collected sum of squares. The fixed coefficient S is recorded by exact exponent notation and is not a variable. Separate producers and checkers verify the substitutions on their nonzero domains, the coordinate inventory, every polynomial coefficient and the exact height. These finite certificates accompany the proofs of the algebraic transformations. The source arithmetic inputs are not assertions certified by these computations.

References

[1] *Small-variable polynomial constructions over the rationals*, companion paper, 7 November 2026. https://github.com/JoseBrox/diophantine-z-q_improvements/tree/main/papers/few_variables.

[2] E. Glazer and contributors, *Diophantine definition of the integers in the rationals*, source repository, coefficient baseline `14e8f601fe0d9173d4a5f084aedc3e07066a7688`, 7 November 2026. The odd-product identity, direct targets and asymmetric-depth construction are in `docs/quartic-523-540/addition.tex`; the repository attributes the odd-product contribution to G. Conant, developed with ChatGPT. https://github.com/elliotglazer/diophantine-z-q/tree/14e8f601fe0d9173d4a5f084aedc3e07066a7688.

[3] Z.-W. Sun, *86 unknowns for Hilbert's tenth problem over the rationals*, arXiv:2610.20691v2, 19 October 2026, especially Theorem 4.1 and Sections 10.3–10.6. https://arxiv.org/abs/2610.20691v2.

[4] J. Voight, *Quaternion Algebras*, Graduate Texts in Mathematics 288, Springer, 2021, especially Proposition 14.6.7. https://jvoight.github.io/quat.html.

[5] J. H. Silverman, *The Arithmetic of Elliptic Curves*, second edition, Graduate Texts in Mathematics 106, Springer, 2009. https://doi.org/10.1007/978-0-387-09494-6.